Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 1 of 7 · Question 16 part b · 4 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Question number	Answer	Mark
16(a)	**One** mark for accurate understanding of Damasio et al. (1994). **One** mark for exemplification of how the Damasio et al. (1994) can be used to explain the scenario. For example: • Damasio et al. (1994) found that patients who had damage to the ventromedial area had difficulty in making rational decisions and processing their emotions (1). Therefore, Marvin may also have difficulty in making rational decisions and control his emotions when working with people who are learning languages at the evening school (1). Accept any other appropriate response.	(2)

Question number	Answer	Mark
16(b)	**One** mark for identification of a strength/weakness (maximum **two** marks). **One** mark for justification, which must be linked to the strength/weakness identified (maximum **two** marks). For example: Strength • Marvin has damaged his ventromedial region which is the same as Phineas Gage who reportedly had impairments to his impulse control after his accident so using this case study has validity (1) because the same brain region has been damaged in Marvin and the patients in Damasio et al. (1994) all had impairments in making rational decisions which means the same is likely to happen to Marvin (1). Weakness • Marvin has damaged Broca's area which were undamaged in the case study of Phineas Gage so Marvin is likely to have some different impairments so using this case study is not fully valid (1) as the damage to Broca's area will cause consequences that Damasio et al. (1994) would not be able to fully predict as their patients did not suffer damage to this region (1). Accept any other appropriate response.	(4)

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 1 of 7 · Question 16 part b · 4 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 2 of 7 · Question 18 part a · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Question number	Answer	Mark
18(a)	**One** mark for accurate understanding of a relevant study selected. **One** mark for exemplification of how the study can be used to explain the scenario. For example: • Damasio et al. (1994) reported that Phineas Gage had become irreverent and capricious after his pre-frontal cortex had been damaged (1), so it is likely that Holly will find that the ten male participants will have become more disrespectful and unpredictable after their pre-frontal cortex damage (1). Accept any other appropriate response.	**(2)**

Question number	Answer	Mark
18(b)	**One** mark for identification of each relevant improvement (maximum **two** marks). **One** mark for justification of each improvement, which must be linked to the improvement identified (maximum **two** marks). For example: • Holly could have asked female participants as well as male participants (1) so that the findings from her study regarding damage to the pre-frontal cortex could be representative of both sexes and not just to men (1). • Holly could have asked the participants' family or peers about their differences in personality before and after the damage (1) because the participant with a damaged brain may not be accurate in their account of the differences in personality whereas their peers or family may be more objective (1). Accept any other appropriate response.	**(4)**

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 2 of 7 · Question 18 part a · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Section E
Social influence – How do others affect you?

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 3 of 7 · Question 13 part a · 1 mark

2020 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Question number	Answer	Mark
13(a)	**One** mark for understanding of an aim of Damasio et al. (1994). For example: • To attempt to locate the areas of damage in Phineas Gage using modern neuroimaging techniques (1). Accept any other appropriate response.	(1)

Question number	Answer	Mark
13(b)	**One** mark for understanding of a part of the procedure used in Damasio et al. (1994). For example: • The researchers used a coordinate system (Talairach's stereotactic space) to map out Phineas Gage's skull (1). Accept any other appropriate response.	(1)

Question number	Answer	Mark
13(c)	**One** mark for each finding identified **A** - Broca's area was undamaged **C** - The ventromedial region of the frontal lobes were damaged The only correct answers are A and C. B is incorrect because the motor cortices were undamaged. D is incorrect because some of the damage was in the frontal lobes. E is incorrect because the dorsolateral region of the front lobes were undamaged.	(2)

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 3 of 7 · Question 13 part a · 1 mark

2020 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 4 of 7 · Question 13 part b · 1 mark

2020 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Question number	Answer	Mark
13(a)	**One** mark for understanding of an aim of Damasio et al. (1994). For example: • To attempt to locate the areas of damage in Phineas Gage using modern neuroimaging techniques (1). Accept any other appropriate response.	(1)

Question number	Answer	Mark
13(b)	**One** mark for understanding of a part of the procedure used in Damasio et al. (1994). For example: • The researchers used a coordinate system (Talairach's stereotactic space) to map out Phineas Gage's skull (1). Accept any other appropriate response.	(1)

Question number	Answer	Mark
13(c)	**One** mark for each finding identified **A** - Broca's area was undamaged **C** - The ventromedial region of the frontal lobes were damaged The only correct answers are A and C. B is incorrect because the motor cortices were undamaged. D is incorrect because some of the damage was in the frontal lobes. E is incorrect because the dorsolateral region of the front lobes were undamaged.	(2)

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 4 of 7 · Question 13 part b · 1 mark

2020 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 5 of 7 · Question 13 part c · 2 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

Question number	Answer	Mark
13(a)	**One** mark for understanding of an aim of Damasio et al. (1994). For example: • To attempt to locate the areas of damage in Phineas Gage using modern neuroimaging techniques (1). Accept any other appropriate response.	(1)

Question number	Answer	Mark
13(b)	**One** mark for understanding of a part of the procedure used in Damasio et al. (1994). For example: • The researchers used a coordinate system (Talairach's stereotactic space) to map out Phineas Gage's skull (1). Accept any other appropriate response.	(1)

Question number	Answer	Mark
13(c)	**One** mark for each finding identified **A** - Broca's area was undamaged **C** - The ventromedial region of the frontal lobes were damaged The only correct answers are A and C. B is incorrect because the motor cortices were undamaged. D is incorrect because some of the damage was in the frontal lobes. E is incorrect because the dorsolateral region of the front lobes were undamaged.	(2)

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 5 of 7 · Question 13 part c · 2 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 6 of 7 · Question 18 part whole · 4 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

Question number	Answer	Mark
18	**One** mark for identification of the strength / weakness (maximum **two** marks). **One** mark for justification, which must be linked to the strength / weakness identified (maximum **two** marks). For example: Strength • Allan has damage in the ventromedial region which is the same as Phineas Gage who reportedly struggled with impulse control so the prediction has validity (1) because the same brain region has been damaged in Allan and the patients in Damasio et al. (1994) all had difficulty in making rational decisions which means the same is likely to happen to Allan (1). Weakness • Allan has damaged Broca's area which was undamaged in Phineas Gage so Allan is also likely to have additional problems so there is a lack of validity to the prognosis (1) because the damage to Broca's area will cause consequences for Allan that Damasio et al. (1994) will not be able to predict as the patients did not suffer from damage to this area (1). Accept any other appropriate response.	**(4)**

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 6 of 7 · Question 18 part whole · 4 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 7 of 7 · Question 16 part a · 2 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Question number	Answer	Mark
16(a)	**One** mark for accurate understanding of Damasio et al. (1994). **One** mark for exemplification of how the Damasio et al. (1994) can be used to explain the scenario. For example: • Damasio et al. (1994) found that patients who had damage to the ventromedial area had difficulty in making rational decisions and processing their emotions (1). Therefore, Marvin may also have difficulty in making rational decisions and control his emotions when working with people who are learning languages at the evening school (1). Accept any other appropriate response.	**(2)**

Question number	Answer	Mark
16(b)	**One** mark for identification of a strength/weakness (maximum **two** marks). **One** mark for justification, which must be linked to the strength/weakness identified (maximum **two** marks). For example: Strength • Marvin has damaged his ventromedial region which is the same as Phineas Gage who reportedly had impairments to his impulse control after his accident so using this case study has validity (1) because the same brain region has been damaged in Marvin and the patients in Damasio et al. (1994) all had impairments in making rational decisions which means the same is likely to happen to Marvin (1). Weakness • Marvin has damaged Broca's area which were undamaged in the case study of Phineas Gage so Marvin is likely to have some different impairments so using this case study is not fully valid (1) as the damage to Broca's area will cause consequences that Damasio et al. (1994) would not be able to fully predict as their patients did not suffer damage to this region (1). Accept any other appropriate response.	**(4)**

Type MS · Damasio et al. (1994): the return of Phineas Gage

Answer 7 of 7 · Question 16 part a · 2 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract